SWIGGY SALES ANALYSIS

SQL Data Analytics Project Report

Restaurant listings • Menu pricing • Food categories

Customer ratings • City-wise patterns

197,430	993	28	28
Dataset records	Unique restaurants	Cities	States

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Technology: Microsoft SQL Server

Dataset period: 01 January 2025 – 31 August 2025

Portfolio case study based on the supplied Swiggy dataset and project presentation.

1. Executive Summary

This project uses SQL Server to explore a supplied dataset of Swiggy restaurant listings and menu information across Indian cities. The analysis focuses on restaurant representation, location coverage, food categories, listed menu prices, displayed customer ratings and date-based record patterns.

The supplied presentation reports 197,430 records, 993 unique restaurant names, 28 cities and 28 states. Key reported measures include an average listed item price of ₹268.51 and an average displayed rating of 4.34 out of 5. These findings describe the supplied dataset; they should not be interpreted as verified order volumes or sales performance.

Report Scope:

The report summarizes the project purpose, dataset, SQL workflow, dimensional model, key findings, interpretation limits and recommended next steps. It does not claim transaction-level sales, customer-level behavior or revenue measurement because those fields are not present in the supplied dataset description.

2. Business Problem

Swiggy has restaurant and food-related records across multiple Indian cities. Reviewing restaurant distribution, menu prices, customer ratings and food-category patterns can help describe the range of restaurant and menu options represented in the dataset.

The central business question is: What do restaurant listings, menu prices and customer ratings reveal about food options and restaurant presence across different locations?

Project Objectives:

- Measure restaurant coverage by city, state and locality.
- Compare menu prices across categories and dishes.
- Identify restaurants and categories with strong listing presence.
- Understand rating distribution and rating-count patterns.
- Generate insights that can guide further business investigation.

3. Dataset Description

Dataset Attribute	Reported Details
Source file	Swiggy_Data.csv
Records	197,430 rows
Columns	10
Unique restaurants	993
Geographic coverage	28 cities and 28 states
Distinct category labels	4,972
Date period	01 Jan 2025 – 31 Aug 2025

Dataset Fields:

Column	Description
State	State name
City	City name
Order_Date	Date field
Restaurant_Name	Restaurant listing name
Location	Local area
Category	Food/menu category
Dish_Name	Listed food item
Price_INR	Listed menu price in rupees
Rating	Displayed customer rating
Rating_Count	Number of ratings

4. Methodology & SQL Workflow

The presentation describes a SQL-led workflow covering data validation, duplicate handling, dimensional modelling, aggregations, date analysis and rankings.

4.1 Data Quality Checks:

The project checks NULL values and blank strings across the ten listed fields. The presentation reports zero NULL cells in the CSV and identifies 27 exact duplicate rows. It also notes that duplicate records should be reviewed before deletion because repeated menu records may not always be erroneous.

Validation Item	Reported Result
NULL cells in CSV	0
Exact duplicate rows	27
Lowest listed price	₹1
Highest listed price	₹8,000

4.2 Duplicate Handling:

The supplied presentation describes duplicate detection using business columns and the use of ROW_NUMBER() to retain one row per duplicate group. Duplicate removal should be performed only after confirming that the rows represent unintended duplicates.

4.3 Dimensional Modelling:

The project presentation describes a Star Schema with a central fact table and five descriptive dimension tables.

Model Component	Fields / Purpose
FACT_SWIGGY_ORDERS	Price_INR, Rating, Rating_Count and dimension keys
DIM_DATE	Full date, year, month, quarter and week
DIM_LOCATION	State, city and location
DIM_RESTAURANT	Restaurant_Name
DIM_CATEGORY	Category
DIM_DISH	Dish_Name

The stated purpose of this model is to reduce repeated descriptive data and support structured analysis by location, date, restaurant, category and dish.

4.4 Analysis Areas:

The project covers record counts and restaurant counts, city and restaurant rankings, food category distribution, menu-price bands, rating distribution and monthly record counts.

5. Key Performance Indicators

KPI	Reported Value	Meaning in This Dataset
Dataset records	197,430	Rows in the supplied dataset
Unique restaurants	993	Distinct restaurant names
Average listed item price	₹268.51	Mean of listed menu prices
Average displayed rating	4.34 / 5	Mean of displayed ratings
Rating count sum	5,591,574	Sum of source rating-count values
Distinct cities	28	Geographic coverage

Metric interpretation: The source data does not include a transaction ID or quantity sold. Therefore, row counts are not confirmed order counts, and SUM(Price_INR) is not verified sales revenue. Treat these as dataset-record and listed-price measures only.

6. Key Findings

6.1 Location Analysis:

Bengaluru has the highest reported record volume, with 20,077 records. The presentation reports coverage across 28 cities and 28 states. This indicates the geographic breadth of the supplied records, not necessarily Swiggy's total market coverage.

City	Records
Bengaluru	20,077
Mumbai	10,507
Hyderabad	10,309
Jaipur	10,286
Lucknow	10,192
New Delhi	10,191

6.2 Restaurant Analysis:

McDonald's is the most represented restaurant name in the reported ranking, with 13,530 records, followed by KFC with 12,961 records. Record frequency indicates how often a restaurant name appears in the dataset; it does not establish order volume, market share or revenue.

Restaurant	Records
McDonald's	13,530
KFC	12,961
Burger King	7,116
Pizza Hut	6,529
Domino's Pizza	5,492
LunchBox - Meals and Thalis	4,700

6.3 Menu Pricing Analysis:

The reported average listed price is ₹268.51 and the median is ₹229. The ₹100–₹199 price band has the largest reported record count, at 56,375, followed by ₹200–₹299 with 54,779 records.

Listed Price Band	Records
Under ₹100	26,803
₹100–₹199	56,375
₹200–₹299	54,779
₹300–₹499	43,773
₹500+	15,700

6.4 Customer Rating Analysis:

The average displayed rating is 4.34 out of 5. The 4.1–4.5 rating band contains the largest reported number of records, at 125,136. Rating counts are source-provided values and should not be interpreted as unique customers without additional data.

Rating Band	Records
≤ 3.0	4,178
3.1–3.5	6,200
3.6–4.0	18,057
4.1–4.5	125,136
4.6–5.0	43,859

6.5 Monthly Record Pattern:

The presentation reports monthly record counts from January to August 2025. Counts range from 23,296 in February to 25,398 in January. These are rows grouped by the recorded date, not confirmed completed orders.

Month	Records	Month	Records
January	25,398	May	25,190
February	23,296	June	24,385
March	24,402	July	24,940
April	24,588	August	25,231

7. Business Interpretation

The analysis provides a descriptive view of restaurant and menu records across the dataset. The location results identify where the dataset contains more records; restaurant rankings show which names appear most frequently; pricing bands describe the distribution of listed item prices; and rating bands summarize displayed customer ratings.

These findings can help frame further questions about restaurant coverage, menu assortment, pricing and customer feedback. However, they cannot establish sales performance, customer demand, profitability or operational efficiency without transaction-level and operational data.

8. Recommendations & Next Steps

1. Validate duplicate records: Review the 27 identified exact duplicates and confirm whether they are unintended repeated rows before removing them.
2. Improve the data model: Validate dimension keys and relationships, and document the grain of the fact table so that each metric has a clear definition.
3. Extend the dataset: Add transaction IDs, quantities, order values, delivery status and customer identifiers (where appropriately governed) to enable actual order and revenue analysis.
4. Investigate location patterns: Compare record coverage by city and locality, while distinguishing listing density from market share.
5. Explore pricing and ratings together: Examine whether listed price bands vary with displayed ratings, category and location. Avoid making causal claims from descriptive comparisons.
6. Build a dashboard: Present record coverage, restaurant representation, price bands, rating distribution and monthly record counts with clear metric definitions.

9. Conclusion

This SQL project demonstrates the use of data validation, duplicate detection, dimensional modelling, aggregations, rankings and date-based analysis to explore a large restaurant and menu dataset. The analysis highlights the distribution of restaurant records, listed prices and displayed ratings across the locations represented in the supplied data.

The project provides a useful foundation for descriptive restaurant and menu intelligence. A future version enriched with transaction-level data could support more direct analysis of orders, sales, customer behavior and business performance.

10. Project Files

Folder / File	Contents
01_Business_Problem/	Business problem PDF
02_Dataset/	Swiggy dataset file
03_SQL_Queries/	SQL Server analysis script
04_Project_Presentation/	Project PowerPoint presentation
05_Project_Screenshots/	Selected project visuals

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